

IP-SAKTI SAHAYAK (IP ██████████ ██████████)

Ayurveda & Traditional Knowledge IP & Regulatory Intelligence Platform
Complete Technical Guide, System Architecture & Operational Blueprint

PROJECT TYPE	TECH STACK	DEPLOYMENT
GovTech / AI LegalTech	React 18, FastAPI, Supabase, BM25 RAG	Vercel (FE), Render (BE), Supabase (DB)

1. Executive Summary & Problem Statement

IP-SAKTI Sahayak (IP ██████████ ██████████) is an AI-powered statutory intelligence and Case Builder portal tailored for Indian traditional medicine innovators (Ayurveda, Siddha, Unani - ASU), biotechnology researchers, startups, and MSMEs. It bridges the gap between grassroots innovation and statutory compliance under Indian Patent Law, the Biological Diversity Act 2002, and TKDL prior-art barriers.

Critical Problems Addressed:

- Section 3(p) Patent Ineligibility:** Section 3(p) of the Indian Patents Act, 1970 bars patenting of traditional knowledge or simple aggregations. Innovators struggle to establish synergy (Section 3(e)) or novel extraction methods without expert guidance.
- Biodiversity Act (BDA 2002) & NBA ABS Clearance:** Section 6 of the Biological Diversity Act 2002 mandates National Biodiversity Authority (NBA) prior approval (Form III) before applying for IP based on Indian biological resources. Violations attract criminal liabilities under Section 55.
- TKDL Prior-Art Conflicts:** India's Traditional Knowledge Digital Library (TKDL) contains 250,000+ formulations. Innovators lack accessible search tools to verify whether formulations already exist in ancient classical treatises.
- Linguistic and Financial Barriers:** Grassroots practitioners and rural MSMEs lack English fluency and cannot afford expensive IP attorneys for preliminary clearance.

■ **SOLUTION:** Provides an AI 5-step Case Builder, grounded Statutory RAG with 4,016 legal corpus chunks, automated Form I/III ABS roadmaps, 100% multilingual localization (Hindi, Marathi, Gujarati, English), and licensed Expert review.

2. Complete Technology Stack

Component Layer	Technologies Used	Key Capabilities
Frontend UI / UX	React 18.3.1, TypeScript, Vite 6.2, Vanilla CSS Glassmorphism, Framer Motion, Lucide Icons	Responsive SPA, glassmorphic dark/light UI, avatar menu, Recharts analytics, and dynamic forms.
State & i18n	Zustand 5.0, TanStack Query v5, i18next & react-i18next, React Hook Form, Zod	Centralized auth/case stores, zero-latency caching, 100% multilingual switching (EN, HI, MR, GU).
Backend API	Python 3.10+, FastAPI 0.110.0, Uvicorn ASGI, Pydantic v2, JWT Auth	Async REST API endpoints, JWT security, CORS policies, case escalation, and health probes.
RAG & Search	BM25 Index (rank-bm25), 4,016 Corpus Chunks, Reciprocal Rank Fusion (RRF), Cross-Encoder	Indexes Patents Act 1970, Patent Rules 2003, BDA 2002, AYUSH guidelines with 4-level legal authority weights.
LLM & AI Engine	Ollama (Llama 3.1 Local), Google Gemini, Groq, OpenAI API fallback	Grounded statutory synthesis, confidence scoring (High/Medium/Low), hallucination elimination.
Database & Auth	Supabase PostgreSQL, Row-Level Security (RLS), uuid-oss, Supabase Auth	User/Expert profiles, multi-turn chat, 5-step case builder submissions, and immutable audit logs.
Cloud Hosting	Vercel (Frontend), Render (FastAPI Docker), Supabase (Managed Postgres)	Global CDN edge delivery, automated CI/CD Git builds, SPA rewrite routing, containerized microservices.

3. System Architecture & Development Methodology

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The architecture features a **Dual-Engine Pipeline**: deterministic Statutory Authority Retrieval combined with an LLM Generation Engine and Human-in-the-Loop Expert Validation.

Hierarchical Architectural Layers:

- **Layer 1: Presentation & Localization Layer:** Vite/React client delivering Citizen & Expert portals, dynamic case wizards, and 4-language i18n support.
- **Layer 2: API Gateway & Service Orchestrator:** FastAPI microservice managing JWT auth, statutory query routing, and Supabase integration.
- **Layer 3: Statutory Knowledge Retrieval Engine:** BM25 index across 4,016 legal chunks with 4-level authority weighting (Acts > Rules > Guidelines > Notifications).
- **Layer 4: Grounded LLM Generation & Confidence Engine:** Synthesizes statutory advice with citation links, abstaining when legal precedent is insufficient.
- **Layer 5: Persistence & Security Layer:** Supabase PostgreSQL with RLS policies isolating user dossiers while allowing assigned expert reviews.

Chronological Development Sequence:

- ◆ **Phase 1: Legal Corpus Ingestion:** Parsed 4,016 chunks from Patents Act 1970, BDA 2002, AYUSH guidelines into structured JSON schemas.
- ◆ **Phase 2: Hybrid RAG Search Engine:** Engineered BM25 index, RRF reciprocal fusion, and hierarchical authority ranking.
- ◆ **Phase 3: Database & Security Setup:** Created Supabase PostgreSQL schema with 10 tables, RLS security policies, and indexes.
- ◆ **Phase 4: FastAPI REST API Handlers:** Implemented async endpoints for `/query`, `/cases`, `/escalate`, and `/expert`.
- ◆ **Phase 5: React Frontend & Case Builder:** Built 5-step wizard, interactive ask portal, and responsive glassmorphic UI.
- ◆ **Phase 6: Multilingual i18n & Auth Hardening:** Added full English, Hindi, Marathi, Gujarati dictionaries and decoupled login state.
- ◆ **Phase 7: Cloud CI/CD & Live Verification:** Deployed frontend to Vercel and backend to Render; verified live end-to-end flows.

4. Core Functional Modules & Features

- **1. Intelligent Ask & Answer Portal (/ask, /answer):** Natural language statutory search returning grounded advice, confidence levels, and exact Act/Section citations.
- **2. 5-Step Guided Case Builder (/case-builder):** End-to-end wizard analyzing formulation novelty, biological resources, Section 3(p) risks, export markets, and ABS clearances.
- **3. Section 3(p) TK Diagnostic Tool (/classifier):** Tests formulations against traditional knowledge exclusions and recommends synergistic claim strategies.
- **4. Prior-Art & TKDL Search (/prior-art):** Screens traditional formulations against classical texts and patent prior-art databases.
- **5. NBA ABS Compliance Navigator (/abs):** Determines mandatory Form I / Form III filings under Biological Diversity Act 2002.
- **6. Expert Review Portal (/expert/dashboard):** Dedicated portal for licensed AYUSH IP attorneys to review, modify opinions, and endorse dossiers.
- **7. Citizen Dashboard & Analytics (/dashboard):** Visual metrics, case progression timelines, notification inbox, and quick-action shortcuts.

5. Database Schema & Supabase Architecture

Table Name	Key Schema Columns	Purpose
profiles	id (UUID PK), email, full_name, role (USER/EXPERT/ADMIN)	Master user and expert identity table.
cases	id (TEXT PK), user_id (FK), title, status, expert_id	Primary case tracker with lifecycle status.
case_builder_data	case_id (FK), biological_material, tk_involved, ingredients	Stores complete formulation and applicant data.
ai_answers	id (UUID), case_id (FK), answer_text, confidence_score	Grounded AI answers with confidence scores.
answer_citations	ai_answer_id (FK), section, act, snippet, authority_level	Verifiable statutory citations linked to answers.
case_events	case_id (FK), event_type, title, description, created_at	Immutable chronological timeline of case actions.
expert_reviews	case_id (FK), expert_id (FK), action, expert_opinion	Expert endorsements, modifications, and legal basis.
notifications	user_id (FK), case_id (FK), title, message, read	In-app real-time notification queue.

6. REST API Endpoints Specification

Method & Route	Payload / Params	Functionality & Response
GET /health	None	Health check & BM25 corpus readiness (4,016 chunks).
POST /api/v1/query	{ question, language }	Statutory RAG query returning grounded answer & citations.
POST /api/v1/cases	{ case_data, ingredients }	Submits Case Builder formulation and triggers AI assessment.
GET /api/v1/cases	Bearer Auth Token	Fetches all active cases belonging to the authenticated citizen.
POST /cases/{id}/escalate	{ reason, domain, notes }	Escalates case to expert pool and updates status to ASSIGNED.
GET /expert/cases	Bearer Auth Token (Expert)	Lists pending review requests for expert evaluation.
POST /expert/cases/{id}/review	{ action, opinion }	Records expert review, updates status, and notifies citizen.

7. Step-by-Step User Journey & Workflow

Citizen / Innovator Workflow:

- **Step 1: Sign Up / Login:** Citizen accesses live portal at <https://ip-shakti-sayahak.vercel.app> and logs in without friction.
- **Step 2: Language & Dashboard:** Selects preferred language (EN, HI, MR, GU). Views key metrics and initiates Case Builder.
- **Step 3: 5-Step Case Formulation:** Fills invention specifics: herbal ingredients, traditional claims, extraction method, export markets.
- **Step 4: Grounded AI Report:** Receives statutory diagnosis: Section 3(p) risk, Section 3(e) synergy requirements, and Form III ABS obligation.
- **Step 5: Expert Escalation:** Submits dossier for review by a certified AYUSH patent attorney.
- **Step 6: Dossier Download:** Receives expert-endorsed filing dossier and proceeds to file with the Indian Patent Office.

Expert Workflow:

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- **Step 1: Expert Portal Login:** Attorney logs into expert dashboard with verified expert credentials.
- **Step 2: Case Inspection:** Examines citizen's formulation, AI statutory citations, and Section 3(p) analysis.
- **Step 3: Legal Endorsement / Modification:** Submits legally binding opinion, recommends claim amendments, and approves dossier.
- **Step 4: Citizen Notification:** System updates case status to REVIEW_COMPLETED and alerts the citizen in real-time.

■ **KEY INNOVATION:** 100% statutory source grounding eliminates hallucination, while 4-language i18n and human expert validation bridge the gap for traditional medicine innovators across India.